miscela gas combustibile + O_2 , T_0

in un recipiente rigido pareti adiab.

Dopo combustione $T_1 > T_0$

$Q, W, \Delta U = ?$

$\underline{Q = 0}$ $\underline{\underline{W = 0}}$ $\Rightarrow Q = \Delta U + W$

$\Rightarrow \underline{\underline{\Delta U = Q - W = 0}}$

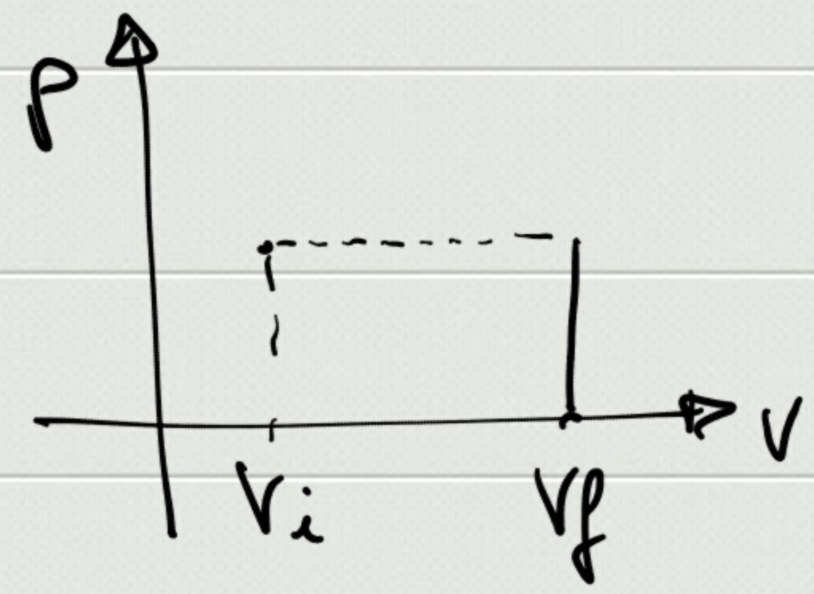
gas espansione $p = 2 \cdot 1.013 \cdot 10^5 \text{ Pa} = \text{cost}$

(isobara)

$V_f = 2V_i$

$Q_{ass} = 5 \cdot 4.186 \cdot 10^3 \text{ J}$

$\Delta U = 4.186 \cdot 10^3 \text{ J}$



$\boxed{V_i = ?}$

$$Q = \Delta U + W \Rightarrow W = Q_{ass} - \Delta U$$

$$\overset{!}{=} \int_{V_i}^{V_f} p(V,T)\,dV = p \int_{V_i}^{V_f} dV =$$

$$= p(V_f - V_i) = pV_i$$

$$V_i = \frac{Q_{ass} - \Delta U}{p} = 0.0826 \text{ m}^3$$

$T_0 = 373.15\ K$

$p = p_{atm} = cost$

$\hookrightarrow 1.013 \cdot 10^5\ Pa$

ΔU (evap. 1g H_2O)

V_{gl} (1g vapor H_2O T_0, p_{atm}) $= 1.67 \cdot 10^{-3}\ m^3$

$$Q = \Delta U + W \Rightarrow \Delta U = Q - W$$

$$\Uparrow$$

$$* \Rightarrow \Delta U = 2090\ J$$

$$Q = m\,\lambda_{ev} \qquad \lambda_{ev,H_2O} = 2.26 \cdot 10^6\ J/kg$$

$$\downarrow$$

$$10^{-3}\ kg \Rightarrow Q = 2.26 \cdot 10^3\ J\ *$$

$$W = \int p\,dV = p \int_{V_i}^{V_f} dV = p\,(V_f - V_i) =$$

$$= 1.013 \cdot 10^5 \left(1.67 \cdot 10^{-3} - 10^{-6}\right) = 169\ J\ *$$